Started on	2024-02-21 18:06
State	Finished
Completed on	2024-02-22 03:06
Time taken	9 hours
Grade	**17.75** out of 20.00 (**89**%)

Question **1**
Correct
Mark 1.00 out of 1.00

Which of these are propositions? What are the truth values of those that are propositions?(Choose NONE if it is not a proposition)

a. Do not go — Not a proposition — NONE

b. What time is it? — Not a proposition — NONE

c. $4 + x = 5$ — Not a proposition — NONE

d. $1 - 2 = 6$ — A proposition — FALSE

e. The moon is made of green cheese — A proposition — FALSE

Question **2**
Complete
Mark 1.00 out of 1.00

Express negation of p and q as English sentences if

p ="There are 13 items in a dozen''

q ="Abby sends more than 100 messages every day''

NOT p = there aren't 13 items in a dozen
NOT q = Abby doesn't send more than 100 messages every day"

Comment:

Question **3**
Complete
Mark 1.00 out of 1.00

i. Express conjunction of p and q as an English sentence if

p =" 3 is larger than 2 ''

q =" 6 times 2 is 10 ''

ii. Is $p \wedge q$ true or false?

i. 3 is larger than 2, and 6 times 2 is 10
ii. false

Comment:

Question 4
Correct
Mark 1.00 out of 1.00

For each sentence determine whether an inclusive or, or exclusive or is intended.

a. Experience with C++ or Java is required [inclusive] ✓

b. Coffee or tea comes with dinner [exclusive] ✓

Question 5
Complete
Mark 1.00 out of 1.00

Write the following proposition in the form ``p if and only if q''.

``It rains if it is a weekend day and it is a weekend day if it rains''

It rains if and only if it is a weekend day

Comment:

Question 6
Partially correct
Mark 0.75 out of 1.00

Determine whether each of these conditional statements is true or false.

a. If $1+1=3$, then unicorns exist [TRUE] ✓

b. If $1+1=2$, then dogs can fly [TRUE] ✗

c. If $2+2=4$, then $1+2=3$ [TRUE] ✓

d. If dogs can fly, then $2+2=4$ [TRUE] ✓

Question 7
Complete
Mark 1.00 out of 1.00

"You can see the movie only if you are over 18 years old or you have the permission of a parent."

Rewrite this sentence using the language of logic in terms of

m="You can see the movie''

e="You are over 18 years old''

p="You have the permission of a parent''

Write your answer on any paper, take a picture and attach it below.

7.jpg

Comment:

Question 8
Complete
Mark 1.00 out of 3.00

"Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded.''

Are these system specifications consistent? Explain your answer.

Write your answer on any paper, take a picture and attach it below.

8.pdf

Comment:

Question 9
Correct
Mark 1.00 out of 1.00

Fill in the following truth table, where p is a proposition.

p	$\neg p$	$p \wedge \neg p$
1	0	0
0	1	0

Question 10
Correct
Mark 2.00 out of 2.00

Fill in the truth table below to show that

$\neg(p \wedge q) \equiv \neg p \vee \neg q.$

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

Question 11
Complete
Mark 3.00 out of 3.00

Using the logical equivalences involving conditional statements show that

$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$

Hint: Use 1, then distributivity, then 1 again.

Write your answer on any paper, take a picture and attach it below.

11.jpg

Comment:

Question 12
Complete
Mark 1.00 out of 1.00

Which normal form has only one true value when $q = 1$ and $p = r = 0$?

Write your answer on any paper, take a picture and attach it below.

12.jpg

Comment:

Question 13
Complete
Mark 3.00 out of 3.00

For a given truth table below construct a compound proposition.

p	q	r	?
1	1	1	0
1	1	0	0
1	0	1	1
1	0	0	1
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	1

Write your answer on any paper, take a picture and attach it below.

13.jpg

Comment: